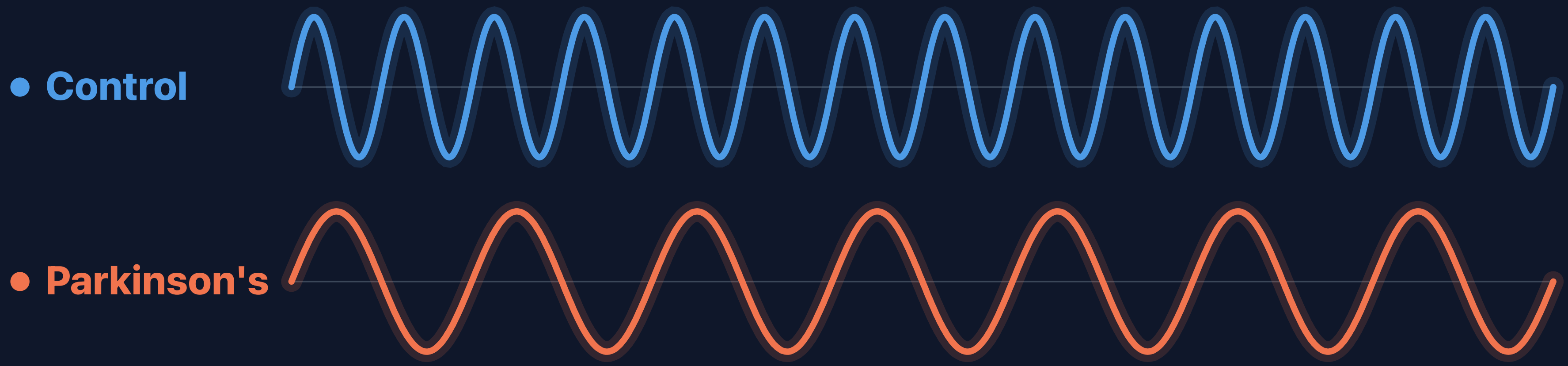
Cortical EEG slowing in resting-state Parkinson's disease

Final presentation: model, code, and findings



PD EEG group · Rui, Jan, Melissa, Friedrich · 26.06.2026

Hypotheses

RESEARCH QUESTION how does resting EEG change in Parkinson's, and can one mechanism explain the slowing?

① theta power higher
in PD · $p < 0.001$

CONFIRMED

② alpha peak slows
9.25 → 8.00 Hz · $p < 0.001$

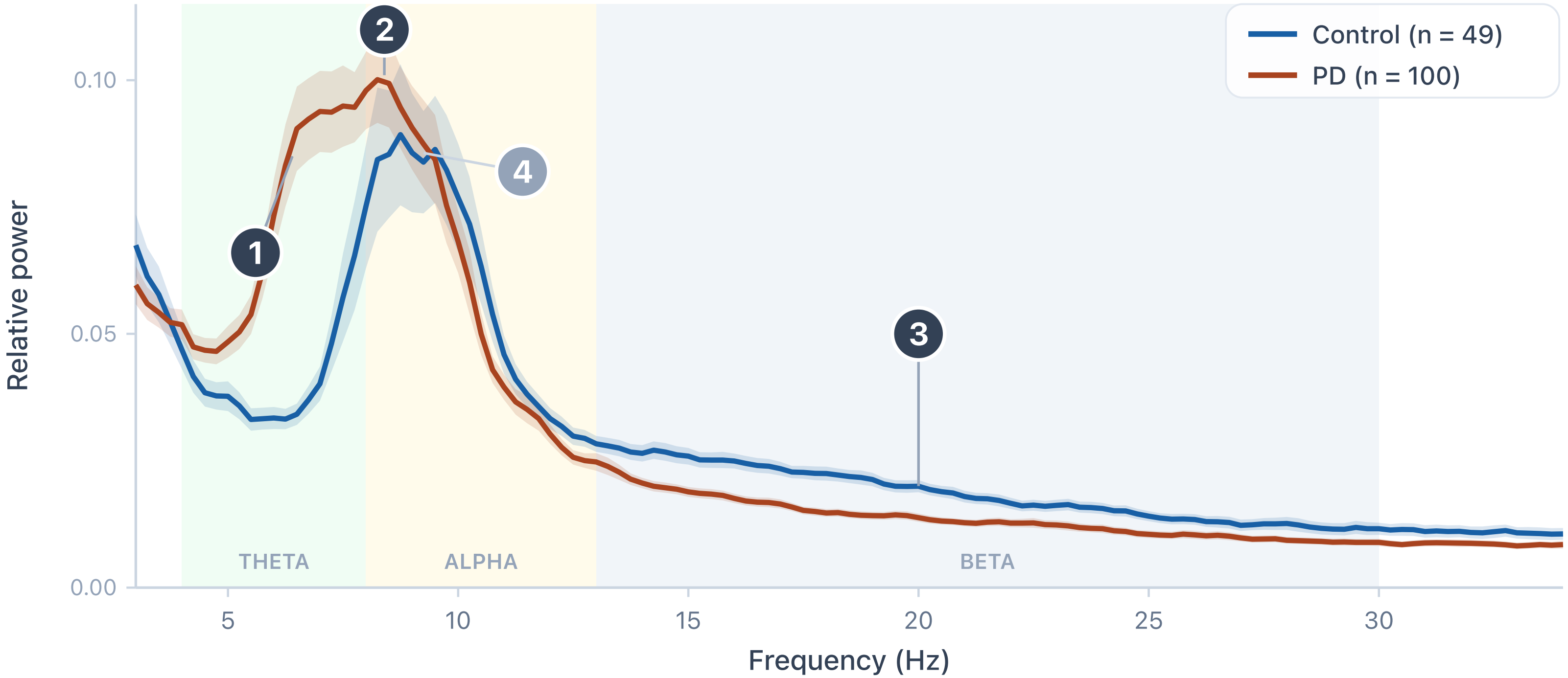
CONFIRMED

③ beta power lower
in PD · $p < 0.001$

CONFIRMED

④ alpha power
unchanged · $p = 0.9$

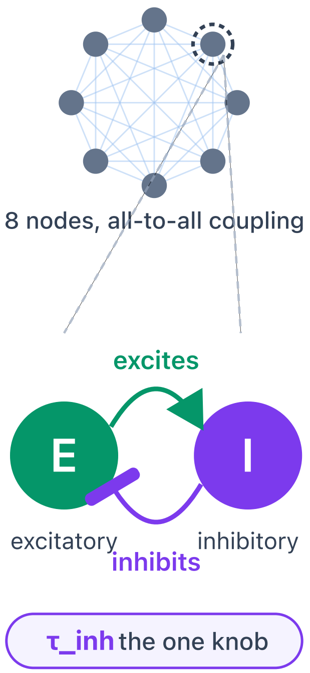
NULL



49 Control, 100 PD · posterior channels · relative power ±SEM · Mann-Whitney U · the three effects hold under Bonferroni (4 tests)

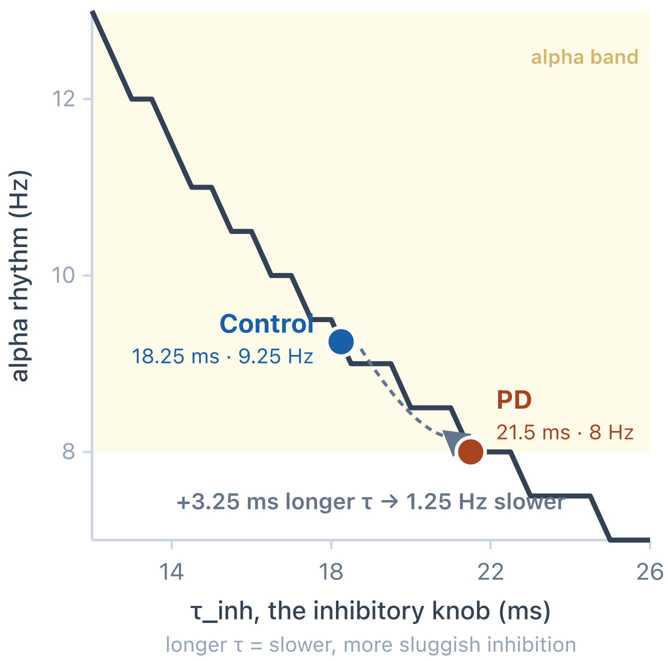
The model: a Wilson-Cowan network

THE MODEL: AN E–I LOOP

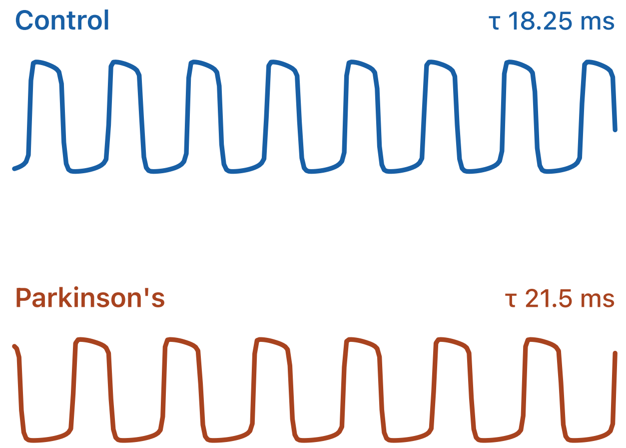


TURN τ_inh UP, THE RHYTHM SLOWS

each point = the model run at one knob setting



THE SIMULATED RHYTHM



same 0.82 s window, the patient rhythm fits in fewer cycles

WHY WILSON-COWAN the rhythm comes out of the excitatory-inhibitory mechanism, so the slowing has a real cause we can name (slower inhibition), not just a frequency we dial down by hand. That cause is the one knob our hypothesis points to.

ONE KNOB ONLY every other setting is held fixed, so the whole change from healthy to patient is a single number, a longer inhibitory time constant **τ_inh** (about **+3.25 ms** in PD).

Code & repository

THE PIPELINE

EEG to result, through six scripts

one number is the whole difference:
the inhibitory time constant **τ_inh**

EEG · 149 → **extract** 01 · features → features.csv → **fit** 03 · per subject → fitted_tau.csv → **compare** 05 · the groups → **PD vs Control**

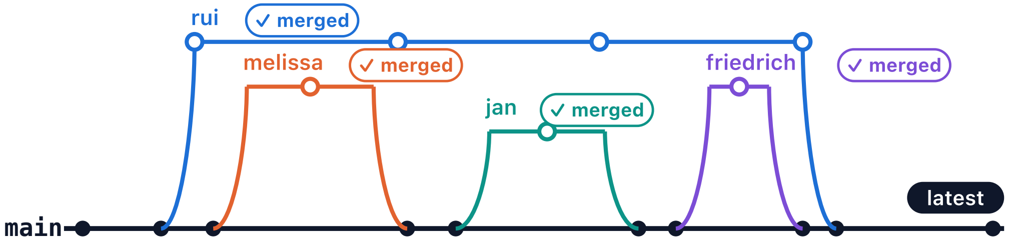
04 data-level test · **02** builds the τ→peak curve · **06** recovers a known τ

THE REPOSITORY

```
pd-eeg                                              main
scripts/                                            01 … 06, numbered
  01_extract_features.py
  02_simulate_network.py
  03_fit_per_subject.py
  04_compare_features.py
  05_compare_tau.py
  06_recover_tau.py
src/                                  features · model · fitting · stats
results/                                       every run writes here
  figures/                     compare_tau.png · tau_recovery.png
slides/                                               decks + notes
```

VERSION CONTROL

64 commits, all merged into main, reproducible



THE MODEL, IN CODE

one function, one free parameter

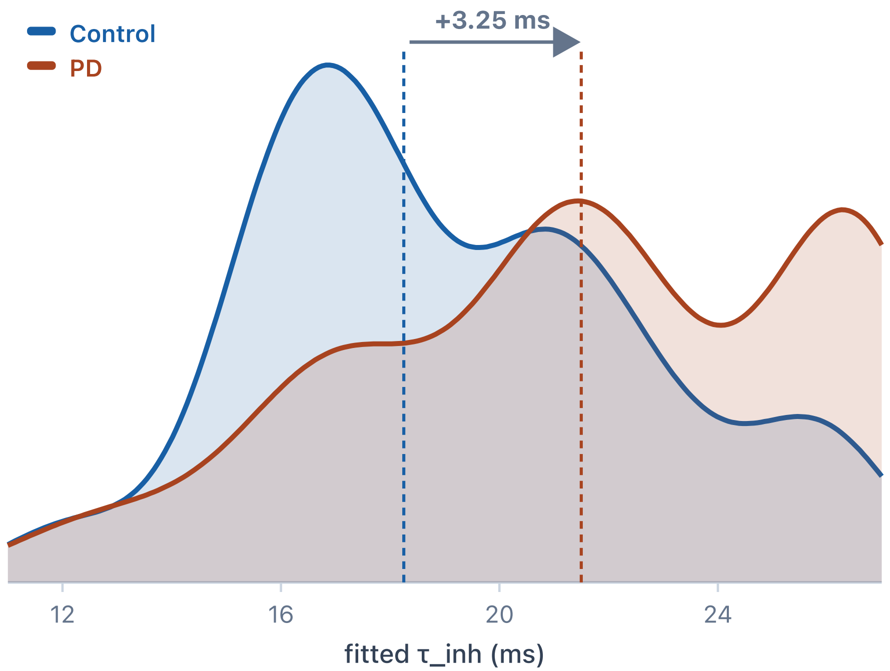
src/model.py

```python
# the whole model is one function; one number is the difference
def build_network(n_nodes, tau_inh, *,
                  exc_ext=0.65, # external drive
                  sigma_ou=0, # deterministic, no noise
                  K_gl=0.6, # global coupling
                  duration_ms=8000): # simulation length
    """A Wilson-Cowan network. tau_inh is the one free
    parameter; everything else stays at these defaults."""
    Cmat, Dmat = all_to_all_coupling(n_nodes)
    model = WCModel(Cmat=Cmat, Dmat=Dmat)
    model.params["tau_inh"] = tau_inh # the one knob we turn
    model.params["exc_ext"] = exc_ext
    model.params["sigma_ou"] = sigma_ou
    model.params["K_gl"] = K_gl
    model.params["duration"] = duration_ms
    return model

# healthy vs patient: same function, one number changed
control = build_network(8, tau_inh=18.25) # alpha 9.25 Hz
patient = build_network(8, tau_inh=21.50) # 8 Hz, +3.25 ms
```

Does the model separate patients from controls?

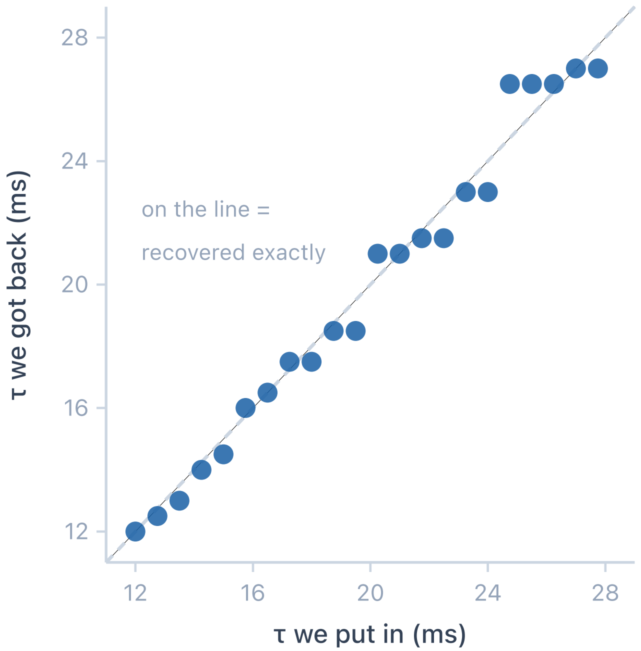
THE SEPARATION fitted τ_inh, Control vs PD



Control 18.25 vs PD 21.5 ms at the median.

The distributions overlap: a random PD is higher ≈ 68% of the time.

PARAMETER RECOVERY put a known τ in, fit it back



The fit is unbiased.

mean error −0.1 ms

typical error ±0.5 ms

Fitted **τ_inh** re-expresses the measured slowing as a model parameter (one-sided Mann-Whitney $p \approx 0.0002$).

Outlook and open challenges

WHERE THE MODEL IS LIMITED

Alpha only
One parameter was tuned to the alpha peak, so theta and beta stay unexplained.

Describes, not proves
A different change could give the same slowing, so it re-describes the data, not the cause.

Coarse and uniform
Frequency is read in coarse steps, on a simplified, all-to-all network.

WHERE IT GOES NEXT

Fit more than the peak
Match the whole spectrum or connectivity, and vary more, to reach theta and beta too.

Anchor it in biology
Tie the fitted inhibition to medication state (on vs off L-Dopa) and symptom severity.

Sharper and more realistic
Fit continuously, not from a lookup table, on a real whole-brain connectome.

What we took away

Our biggest lessons were less about the model than about how a team does computational research well.

REPRODUCIBILITY
Rui

Carrying this end to end, from raw EEG to the fitted model, taught me that a result is only as trustworthy as the pipeline behind it; the reproducible scripts and the honest recovery check mattered as much as the model.

APPROACHABLE
Jan

This course taught me that you do not have to be a master programmer to meaningfully contribute to, and clinically interpret, advanced computational research.

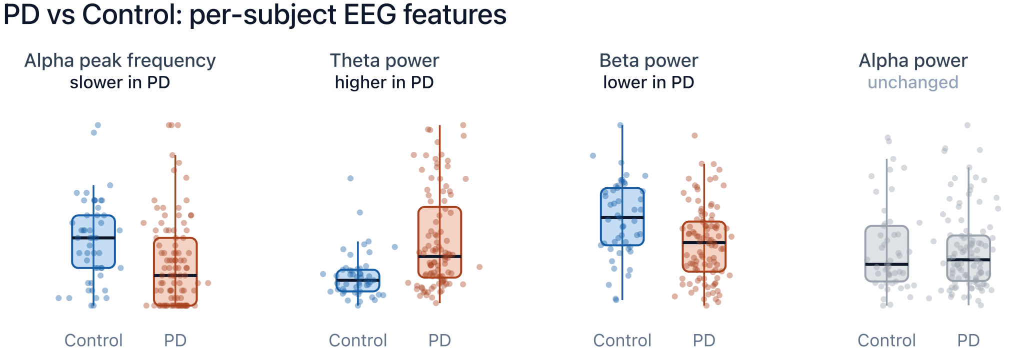
DOCUMENTATION
Melissa

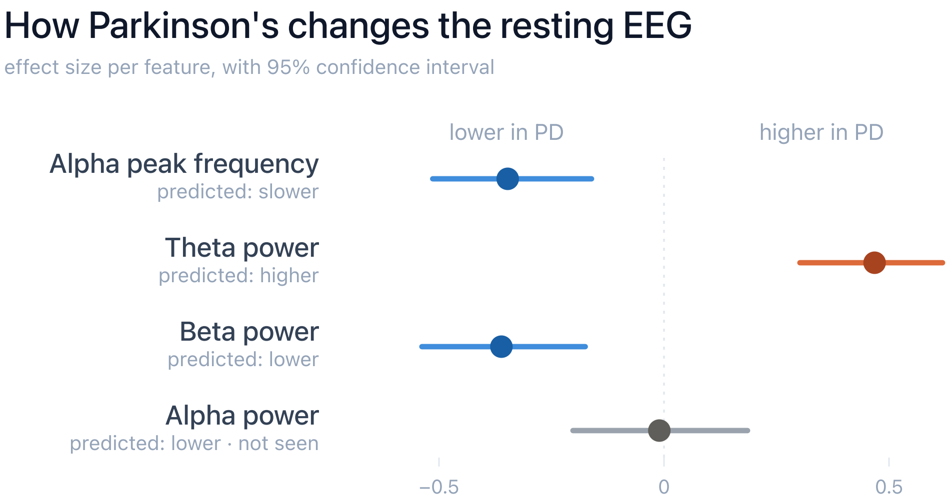
My biggest learning was how much easier good documentation makes everything, whether it's the repository structure, script naming, clear responsibilities, or the code itself. It keeps the whole team aligned, helps everyone track progress, and makes even complex or unfamiliar methods feel more approachable.

COMMUNICATION
Friedrich

How big a difference good communication makes. Even missing one group meeting meant you basically missed all the progress. In-person communication is essential to work well together and establish a group-wide understanding.

Appendix and references





DATA Anjum MF, et al. Resting-state EEG measures cognitive impairment in Parkinson's disease. npj Parkinson's Disease (2024). doi:10.1038/s41531-023-00602-0

MODEL Wilson HR, Cowan JD. Excitatory and inhibitory interactions in localized populations of model neurons. Biophysical Journal 12, 1-24 (1972).

TOOL Cakan C, Jajcay N, Obermayer K. neurolib: a simulation framework for whole-brain neural mass modeling. Cognitive Computation (2023). doi:10.1007/s12559-021-09931-9